Network Threats and Mitigation & Physical Security and Risk

1.Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Network Threat	What it means	Real-world example
Phishing	Fake messages or websites are used to steal passwords or personal information.	WhatsApp: A user may receive a fake message claiming to be from WhatsApp support and asking them to click a link and enter their account details.
DDoS (Distributed Denial of Service)	Attackers send a huge amount of traffic to a server to make the service slow or unavailable.	Instagram: Attackers could flood Instagram-related servers with requests, causing users to face slow loading or temporary service disruption.
Man-in-the-Middle (MITM)	An attacker secretly intercepts communication between two devices or users.	Paytm: On an unsecured public Wi-Fi network, an attacker could try to intercept communication between a user's device and the service.
Malware	Malicious software designed to damage a device or steal information.	WhatsApp: A user may receive a malicious file or link that installs malware when opened, potentially stealing data from the phone.
Password Attack	An attacker tries to guess, steal, or crack a user's password.	Instagram: An attacker may repeatedly try common or leaked passwords to gain unauthorized access to an Instagram account.

**2.Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.**

```
# Simple Brute-Force PIN Simulation


preset_pin = "5832"

attempts = 0


for i in range(10000):
    guess = f"{i:04d}"
    attempts += 1


    if guess == preset_pin:
        print("PIN guessed successfully:", guess)
        print("Number of attempts:", attempts)
        break
```

Output:

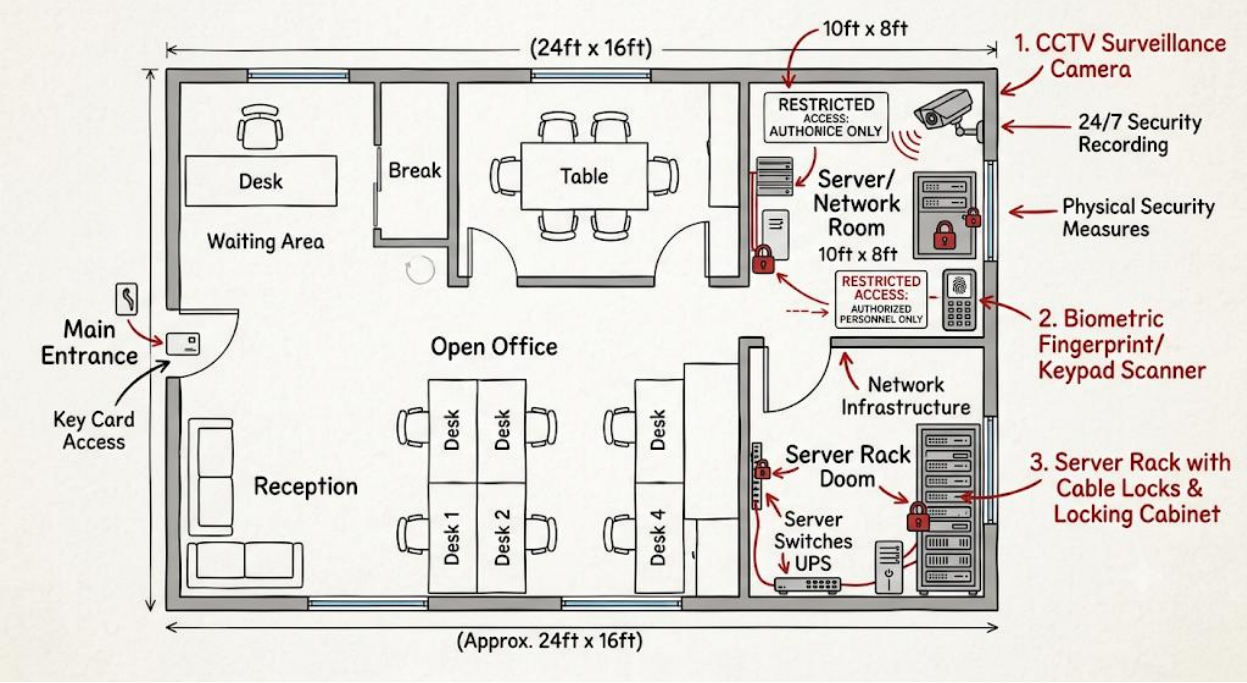
PIN guessed successfully: 5832

Number of attempts: 5833

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Physical Security Risk	Mitigation Step
1. Unauthorized Access	Use biometric access or RFID cards and allow entry only to authorized staff.
2. Fire	Install smoke detectors and an automatic fire-suppression system suitable for server rooms.
3. Water Leakage/Flooding	Keep servers away from water pipes and install water-leak detection sensors.
4. Power Failure	Use UPS systems and backup generators to keep servers running during power outages.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

No.	Action	Purpose
1	Use strong passwords and enable MFA for admin and important accounts.	Prevent unauthorized account access.
2	Keep antivirus and operating systems updated on all lab computers.	Protect against malware and known vulnerabilities.
3	Configure a firewall and secure Wi-Fi using WPA2/WPA3.	Block unauthorized network traffic and access.
4	Regularly back up important lab data to a secure location.	Recover data after malware, accidental deletion, or system failure.
5	Install CCTV and control physical access to the server/network room.	Prevent unauthorized physical access to network equipment.
6	Secure network devices and cables using locked racks and cable locks.	Prevent tampering, theft, or accidental disconnection.
7	Monitor network activity and report suspicious incidents.	Detect threats such as unusual login attempts, malware, or unauthorized devices early.